

Telegram Mini App (TMA) — Frontend Dasturchi Qo'llanmasi

Zero-Click Authentication, Kanban Board API va Telegram Native UX Integratsiyasi

💡 **Asosiy Fikr:** Telegram Mini App ilovangizda foydalanuvchiga *login yoki parol kiritish umuman kerak emas!* Telegram ilova ochilganda xavfsiz `initData` kalitini uzatadi. Frontend ushbu kalitni bir marta backendga beradi va doimiy **JWT Bearer Token** oladi.

1-BOSQICH

SDK yuklash

Telegram WebApp JS SDK yuklanadi va oyna to'liq ekranga kengaytiriladi.

2-BOSQICH

Zero-Click Auth

`initData` backendga yuborilib, 2 soniyada JWT token olinadi.

3-BOSQICH

Doskani olish

Birgina `/board` so'rovi bilan ustunlar va kartochkalar chiziladi.

4-BOSQICH

Telegram UX

MainButton, BackButton va Haptic vibratsiya ulanadi.

1. Telegram WebApp SDK ni Ulash

Loyiha `index.html` faylining `<head>` qismiga rasmiy Telegram skriptini qo'shing:

```
<!-- Telegram Mini App rasmiy JS SDK -->
<script src="https://telegram.org/js/telegram-web-app.js"></script>
```

Ilova ishga tushganda (React `useEffect` yoki ilova startida) oynani tayyorlang:

```
if (window.Telegram?.WebApp) {
  window.Telegram.WebApp.ready();
  window.Telegram.WebApp.expand(); // Mini App'ni to'liq ekranga yoyish
}
```

2. Avtorizatsiya: `initData` orqali JWT Token Olish

Ilova ochilishi bilanq quyidagi so'rov yuboriladi:

METOD	URL	AUTH TALABI
POST	/api/auth/telegram-webapp	Talab qilinmaydi (Public)

Request Body (JSON):

```
{
  "initData": "query_id=AAHdF...&user=%7B%22id%22%3A123456%7D&auth_date=1710000000&hash=d824..."
}
```

Response (200 OK):

```
{
  "success": true,
  "message": "Telegram orqali tizimga muvaffaqiyatli kirdingiz",
  "data": {
    "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...", // ← JWT Bearer Token
    "refreshToken": "48b6fa20-435d-4f11-...",
    "user": {
      "id": "c1f7a08b-b384-4fe1-...",
      "name": "bekmurod_dev",
      "fullName": "Bekmurod",
      "role": "USER"
    }
  }
}
```

Xavfsizlik: Olingan `data.token` ni `localStorage.setItem('token', token)` qilib saqlang va keyingi barcha so'rovlarga `Authorization: Bearer <token>` sifatida qo'shing.

3. Asosiy API Endpoints (CRUD)

Barcha endpointlar uchun `Authorization: Bearer <token>` sarlavhasi berilishi shart.

METOD	ENDPOINT	TAVSIFI
GET	/api/workspaces	Foydalanuvchining barcha loyihalari ro'yxati
GET	/api/workspaces/{id}/board	Kanban doskasi: ustunlar va ulardagi vazifalar
POST	/api/workspaces/{id}/tasks	Yangi vazifa yaratish
PATCH	/api/workspaces/{id}/tasks/{taskId}	Ustunni o'zgartirish (Drag-and-Drop / Move)
DELETE	/api/workspaces/{id}/tasks/{taskId}	Vazifani o'chirish

Kanban Dorkasini Olish Natijasi (`GET /api/workspaces/{id}/board`):

```
[
  {
    "id": "col-todo-id",
    "title": "To Do",
    "order": 0,
    "cards": [
      {
        "id": "task-123",
        "publicId": "TASK-45",
        "title": "Telegram Mini App UI dizayni",
        "priority": "HIGH",           // LOW | MEDIUM | HIGH | URGENT
        "dueDate": "2026-10-15",
        "labels": [{ "name": "Frontend", "color": "#3B82F6" }],
        "assignees": [{ "name": "bekmurod", "fullName": "Bekmurod" }]
      }
    ]
  },
  {
    "id": "col-inprogress-id",
    "title": "In Progress",
    "order": 1,
    "cards": []
  },
  {
    "id": "col-done-id",
    "title": "Done",
    "order": 2,
    "cards": []
  }
]
```

Yangi Vazifa Qo'shish (`POST /api/workspaces/{id}/tasks`):

```
{
  "columnId": "col-todo-id",           // Majburiy
  "title": "Telegram API integratsiyasini sinash", // Majburiy
  "description": "Batafsil ma'lumot matni...",
  "priority": "HIGH",
  "dueDate": "2026-10-20"
}
```

Vazifani Ustunini O'zgartirish (`PATCH /api/workspaces/{id}/tasks/{taskId}`):

```
{
  "columnId": "col-inprogress-id" // Yangi ustun UUID si
}
```

4. Telegram Native UI Imkoniyatlari (Premium UX)

Telegram Mini App foydalanuvchiga tabiiy mobil ilovadek tuyulishi uchun rasmiy funksiyalardan foydalaning:

1. Telegram Ranglariga Moslashish (CSS Variables):

```
body {
  background-color: var(--tg-theme-bg-color, #ffffff);
  color: var(--tg-theme-text-color, #0f172a);
}

.card {
  background-color: var(--tg-theme-secondary-bg-color, #f8fafc);
  border: 1px solid var(--tg-theme-hint-color, #e2e8f0);
}

.btn-primary {
  background-color: var(--tg-theme-button-color, #2563eb);
  color: var(--tg-theme-button-text-color, #ffffff);
}
```

2. MainButton, BackButton va Haptic Feedback:

```
const tg = window.Telegram?.WebApp;

// 1. Pastki yopishib turuvchi asosiy tugma
tg.MainButton.setText("✚ YANGI VAZIFA");
tg.MainButton.show();
tg.MainButton.onClick(() => {
  openCreateModal();
});

// 2. Tepada orqaga qaytish tugmasi
tg.BackButton.show();
tg.BackButton.onClick(() => {
  closeModalOrGoBack();
});

// 3. Harakatlarda haptic vibratsiya (tugma bosilganda / kartochka surilganda)
tg.HapticFeedback.impactOccurred('medium'); // 'light' | 'medium' | 'heavy'
tg.HapticFeedback.notificationOccurred('success'); // 'success' | 'warning' | 'error'
```

5. Frontend Dasturchi Uchun Tayyor `api.js` (Copy-Paste)

```

import axios from 'axios';

const API_BASE = 'https://your-backend-api.com'; // Yoki http://localhost:8080

export const api = axios.create({
  baseURL: API_BASE,
  headers: { 'Content-Type': 'application/json' }
});

// Avtomatik JWT token biriktirish
api.interceptors.request.use((config) => {
  const token = localStorage.getItem('tc_jwt');
  if (token) config.headers.Authorization = `Bearer ${token}`;
  return config;
});

// 1. Telegram Zero-Click Login
export async function loginWithTelegram() {
  const initData = window.Telegram?.WebApp?.initData;
  if (!initData) return null;

  const res = await api.post('/api/auth/telegram-webapp', { initData });
  const { token, user } = res.data.data;
  localStorage.setItem('tc_jwt', token);
  return user;
}

// 2. Kanban Doskasini yuklash
export async function fetchBoard(workspaceId) {
  const res = await api.get(`/api/workspaces/${workspaceId}/board`);
  return res.data.data;
}

// 3. Vazifa ustunini o'zgartirish
export async function moveTaskColumn(workspaceId, taskId, columnId) {
  const res = await api.patch(`/api/workspaces/${workspaceId}/tasks/${taskId}`, { columnId });
  return res.data.data;
}

```

6. BotFather da Mini App Tugmasini Sozlash

3 Oddiy Qadamda Botga Ulash:

1. Telegramda **@BotFather** ga kiring va `/mybots` ni bosing.
2. Botingizni tanlab: **Bot Settings** ➔ **Menu Button** ➔ **Configure menu button** ga bosing.
3. Frontend Mini App havolangizni (masalan: `https://task-center.vercel.app`) va tugma nomini kiriting: `Doska`.

Natija: Bot ichidagi chap pastki burchakda qulay Mini App tugmasi paydo bo'ladi!